

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460313.6269 +/- 0.0034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.051 +/- 0.05
Transit depth (Rp/Rs)^2	0.26 +/- 0.51 [%]
Orbital Inclination (inc)	85.11 +/- 1.72
Airmass coefficient 1 (a1)	52.03 +/- 34.16
Airmass coefficient 2 (a2)	-3.0 +/- 1.48
Scatter in the residuals of the lightcurve fit is	58.76 %
Best Comparison Star	#2 - [206, 308]
Optimal Aperture	2.9
Optimal Annulus	4.77
Transit Duration (day)	0.032 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.051 ± 0.05 vs 0.131 (deviation 1.17σ)
Duration fit vs published	0.032 d vs 0.1075 d (deviation 1.62σ)
Mid-time O–C	0.9 min
Depth SNR	0.7
Residual scatter	58.76 %

Light curve

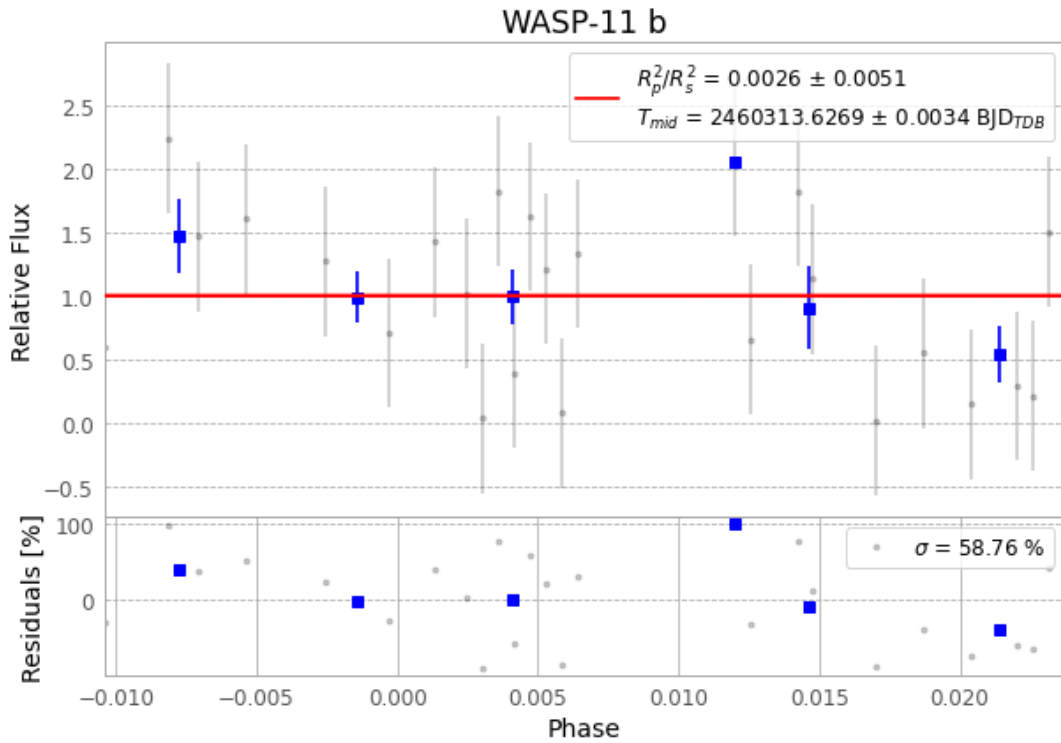


Figure 1 — FinalLightCurve_WASP-11 b_2024-01-03.png (SHA-256 40c88bf7ee1a195a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [477, 129], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ad42415701dd0898...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

65 FITS frames (43 MB), HATP-10240104020015.FITS → HATP-10240104051215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-240104.log (70355ef7f37f...), FinalLightCurve_WASP-11 b_2024-01-03.png (40c88bf7ee1a...), FinalLightCurve_WASP-11 b_2024-01-03.csv (8fba15fbc29a...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 01:38Z.